TaxonVision

Production Biodiversity MLOps Architecture

High-Throughput Vision · Split Conformal Safety · Automated Closed Loop



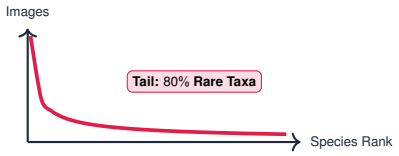
Benjamin Förster

Final MLOps Project Presentation & Architecture Defense

The Domain Challenge

Why Standard Computer Vision Fails in the Wild

Extreme Long-Tail Distribution



High Cost of Hallucinations

Misclassifying an endangered organism as a common species invalidates biodiversity surveys and damages conservation.

Three Production Realities

- **Severe Class Imbalance:** A few common taxa have 10,000+ photos; thousands of rare species have < 10.
- **Continuous Feature Drift:** Lighting, seasonal changes, camera sensors, and geographic shifts constantly distort the visual manifold.
- **Novelty & Noise:** Users submit unseen insects, blurred frames, and blank shots – the model must abstain!

The MLOps Paradigm Shift

From Notebook Prototypes to an Engineered Production System

Prototype / Toy Machine Learning

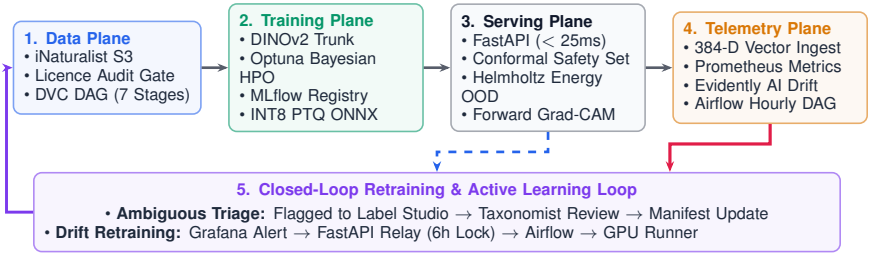
- **Unversioned Data:** Ad-hoc CSVs, non-reproducible manual train/test splits.
- **Redundant Compute:** Heavy vision backbones re-extracted on every run.
- **Raw Softmax Guesses:** Uncalibrated probabilities, forced to predict on unknown classes.
- **Silent Decay:** Zero drift monitoring; model silently fails in production.

TaxonVision Production MLOps

- **DVC Versioned DAG:** 7 pure stages cryptographically tracked by MD5 hashes.
- **Zero-Copy Feature Caching:** Pre-computed foundation embeddings cached on NVMe.
- **Conformal Safety Sets:** Guaranteed coverage ($1 - \alpha \geq 95\%$) with Energy OOD filter.
- **Closed Retraining Loop:** Automated alerts $\rightarrow$ Airflow DAG $\rightarrow$ GPU retraining.

End-to-End System Architecture

The Five Operational Planes of TaxonVision



Data Plane: Ingestion & Licensing Compliance

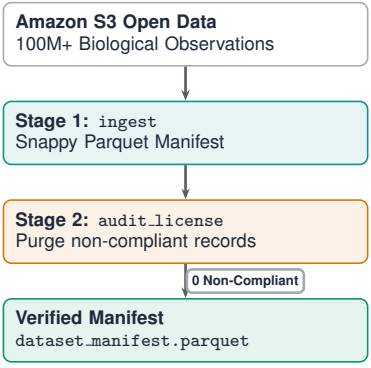
Zero-Copy S3 Streaming and Strict Creative Commons Governance

Zero-Copy Ingestion Flow

- **Amazon S3 Streaming:** Direct manifest and crop streaming from `s3://inaturalist-open-data/`.
- **Apache Arrow IPC:** High-speed in-memory tabular processing with embedded DuckDB filtering.
- **Ephemeral Images:** Raw JPEGs are normalized into tensor memory and discarded – never stored as duplicate disk files.

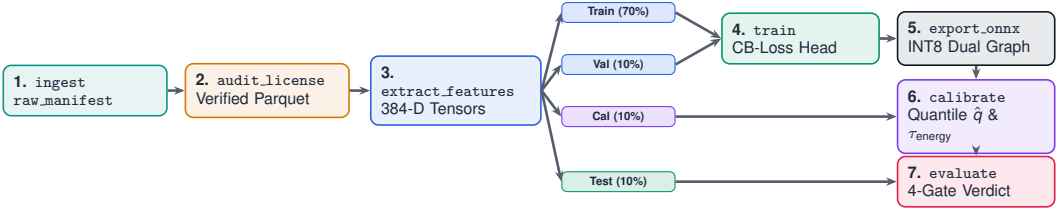
License Compliance Gate (Rule 05)

- **Whitelisted:** `CC0`, `CC-BY`, `CC-BY-NC`.
- **Rejected:** `All-Rights-Reserved`, `CC-BY-ND`.
- **Attribution Invariant:** Photographer credit preserved across all manifest stages and API endpoints.



The Epistemic Diamond DAG

7-Stage Reproducible Pipeline with Zero Split Leakage



Strict Split Isolation (Rule 01): Train updates head weights; **Val** guides Optuna HPO early stopping; **Cal** quantiles conformal coverage; **Test** renders final promotion verdict. Zero leakage!

Dual-Tier Storage Topology

Cloud Collaboration (DagsHub) vs. Local Serving Economics

Tier 1: Cloud Collaboration (DagsHub)

- **Remote:** Hosted DagsHub S3 (`s3://dvc`).
- **Role:** Immutable public archive for manifests, embeddings, and registered ONNX models.
- **CI Integration:** GitHub Actions pulls checkpoints using `DAGSHUB_TOKEN` without cluster access.
- **Open Science:** Public MLflow experiment tracker.

Tier 2: In-Cluster Storage (Workstation)

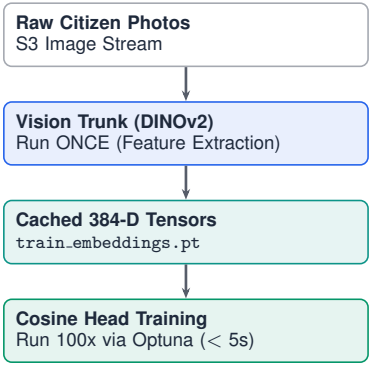
- **Remote:** Local SeaweedFS S3 (`local-seaweedfs`).
- **Role:** High-throughput NVMe-backed storage for embeddings and local model staging.
- **Zero Egress Costs:** 100,000s of feature embeddings synced locally without cloud bandwidth bills.
- **Private MLflow 3.x:** High-frequency Optuna logging.

Model Architecture & Feature Caching

Accelerating Iteration Velocity by 100x

Decoupled Vision Architecture

- **Frozen Trunk:** Foundation vision backbone (DINOv2 ViT-Small, $D = 384$).
- **Lightweight Head:** Cosine-similarity classification head mapping embeddings to species logits.
- **The Feature Caching Strategy:**
  - Forward-passing through the 22M parameter ViT takes minutes.
  - Once pre-computed, visual representations never change!
  - Extract features **once**; cache `.pt` tensors on NVMe.
- **Result:** Head training runs in < 5 **seconds**!



Latency Optimization: Zero-Allocation INT8 ONNX

Deterministic Low-Latency Inference Under Strict Production SLOs

11.8 ms
CPU p95 (Budget: $\leq$ 15 ms)

6.2 ms
GPU p95 (Budget: $\leq$ 10 ms)

18.4 ms
Microservice p95 (Budget: $\leq$ 25 ms)

Dual-Output ONNX Architecture

- Export outputs both classification logits and normalized 384-D feature embeddings in a **single forward pass**.
- Eliminates redundant forward passes for telemetry logging and drift analysis.

Zero-Allocation Discipline (Rule 02)

- Pre-allocated contiguous C-order NumPy arrays for input and prediction buffers.
- Zero Python dictionary boxing or dynamic list allocations inside high-frequency serving loops.

Epistemic Safety: The Problem with Softmax

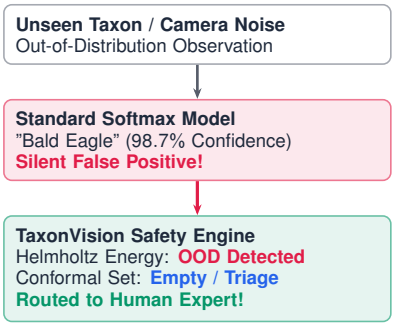
Why Deep Neural Networks Fail Silently in Nature

The Illusion of Softmax "Confidence"

- Softmax mathematically forces probabilities to sum to 1.0 (100%).
- On unseen species, camera glare, or corrupt images, standard models **cannot** say "I don't know".
- Result: High-confidence ($> 95\%$) silent hallucinations on completely wrong taxa!

The Production Safety Imperative

- We require a mathematical safety envelope that rejects false certainty before predictions reach end users.



Split Conformal Prediction Made Simple

Mathematically Guaranteed Coverage Sets Instead of Single Guesses

The Conformal Prediction Guarantee

- Instead of forcing a single guessed species, the model outputs an **adaptive prediction set** $C(X)$.
- The user specifies a safety confidence: e.g. **95**% coverage ($\alpha = 0.05$).
- **Mathematical Guarantee (Rule 06):** The true species is mathematically guaranteed to be in the set $\geq$ 95% of the time:

$$P(Y \in C(X)) \geq 1 - \alpha$$

- Calibrated strictly on the holdout **Calibration split**.

Three Operational Set Outcomes

Case 1: Decisive ($|C| = 1$)
- Set: {`"Barn Owl"`}
- High confidence; automated response.

Case 2: Ambiguous ($1 < |C| \leq 3$)
- Set: {`"Common Gull"`, `"Herring Gull"`}
- Narrows search to sister species.

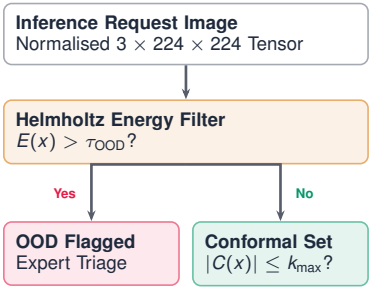
Case 3: Referral ($|C| > 5$ or $|C| = 0$)
- Model defers to human triage queue!

Out-of-Distribution Gating

Helmholtz Energy Filter and Human Referral Boundary

Energy-Based OOD Gating

- Computes negative Helmholtz free energy $E(x)$ over unconstrained model logits.
- In-distribution taxa produce low energy; novel organisms, blurred photos, or blank frames produce high energy.
- Threshold τ_{OOD} calibrated to maintain 95% true positive rate on calibration data.
- Energy $>$ τ_{OOD} $\implies$ Immediate bypass of classification directly to human triage.

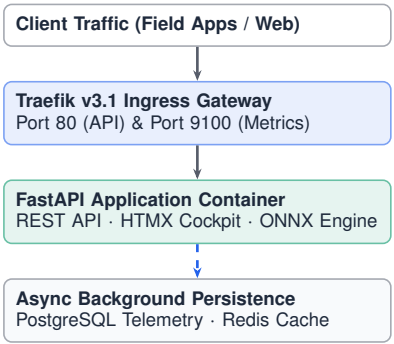


Unified Serving: REST API & HTMX Dashboard

Dual-Interface Microservice Without Frontend Build Ballast

Unified Dual-Interface Architecture

- **Machine REST API:** Strongly typed Pydantic endpoints (`/api/v1/predict`, `/explain`, `/drift/evaluate`).
- **Operator Dashboard:** Server-rendered Jinja2 + HTMX cockpit at `/dashboard`; zero JavaScript build complexity.
- **Forward Grad-CAM Attribution:** Generates visual attribution heatmaps on demand off the event loop thread.
- **Traefik Edge Reverse Proxy:** Handles SSL termination, load balancing, and rate limiting.



Telemetry & Database Feature Persistence

Asynchronous PostgreSQL Ingestion Without Serving Latency Penalty

Why Persist 384-D Embeddings?

- Detecting representation shift requires analysing feature manifolds from live traffic.
- Storing raw citizen photos wastes storage and violates privacy.
- TaxonVision persists the **384-dimensional feature vector** (`embedding REAL[]`) directly in PostgreSQL.
- **Event Loop Protection (Rule 08):** Written asynchronously via `BackgroundTasks` – request returns in 18 ms while I/O runs in background.

Telemetry Schema (`predictions` table)

- `observation_id` (UUID primary key)
- `created_at` (Timestamptz index)
- `top1_taxon` & `top1_confidence`
- `conformal_set` (`TEXT[]` array)
- `ood_energy` & `is_ood` (Boolean)
- `latency_ms` (Measured wall-clock)
- `embedding` **(`REAL[]`, 384 dimensions)**

Ready for instantaneous sliding-window drift queries!

Automated Drift Detection in Practice

Two-Tier Hypothesis Testing and Evidently AI Lineage

Two-Tier Drift Evaluation Engine

- **Tier 1: Wasserstein-1 (W_1) on Norms**
  Fast $O(N)$ linear-time check on embedding lengths; catches immediate camera/contrast shift.
- **Tier 2: Kernel MMD² Permutation Test**
  Multivariate distribution test ($B = 200$ permutations); detects subtle manifold distortion ($p < 0.01$).
- **Evidently AI 0.7 HTML Dashboard:**
  Projects 384 dims to 8 PCA components and renders interactive reports (`drift_report.html`).

Automated MLflow Tracking Lineage

- Evaluated hourly by the Airflow DAG `prediction_drift_monitor`.
- Automatically creates an MLflow run in experiment `taxon-vision-drift`.
- Logs metrics (`drift_w1`, `drift_mmd2`, `p_value`), and tags `drift_detected=true/false`.
- Attaches the Evidently HTML report as an immutable build artifact.

Prometheus & Grafana Observability Cockpit

Real-Time Operational Monitoring and Four Alerting Rules

Real-Time Metrics & Dashboards

- **Prometheus Pull Scraper:** Scrapes `/metrics` every 15 seconds across serving and Traefik pods.
- **Grafana Cockpit (6 Panels):**
  - Request rate and status codes (2xx, 4xx, 5xx).
  - p50, p95, p99 end-to-end latency histograms.
  - Conformal set size distribution.
  - Human triage referral rate.
  - Live drift distance gauge (W_1).

Four Production Alerting Rules

- `PredictionLatencyP95High`: p95 latency $>$ 120 ms for 5 min (`warning`).
- `PredictionErrorRatioHigh`: Error ratio $>$ 5% for 5 min (`critical`).
- `EmbeddingDriftDetected`: Drift gauge $>$ 0.5 (`critical`, **action: retrain**).
- `DriftMonitorStale`: No drift run in $>$ 2 hours (`warning`).

Closed-Loop Retraining Trigger

Grafana Alert Webhook → FastAPI Relay → Airflow 3.0 DAG



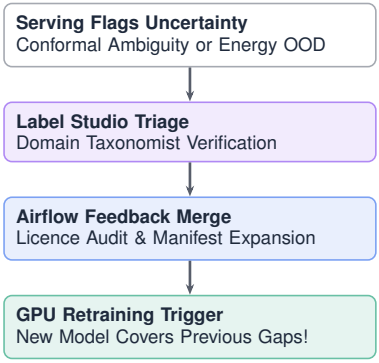
Anti-Alert-Storm Cooldown Protection: When sudden environmental drift causes continuous alerting, the atomic 6-hour cooldown window ensures the GPU cluster executes only one retraining run, preventing resource exhaustion.

Human-in-the-Loop Active Learning

Turning Uncertain Predictions into Retraining Fuel

The Closed Curation Lifecycle

- **1. Epistemic Triage:** Ambiguous sets ($|C| > 3$) and empty sets ($|C| = 0$) are saved to `data/flagged_observations/`.
- **2. Taxonomist Review:** Taxonomists inspect specimen crops and CAM attribution heatmaps in Label Studio.
- **3. Feedback Export:** Label Studio exports verified JSON multi-class annotations.
- **4. Airflow Ingestion:** The `model_retraining` DAG merges feedback, validates open licences, and updates the DVC manifest.



Four-Gate Model Promotion Verdict

Strict Automated Quality Verification Before Production Alias

Why Validation Accuracy Is Not Enough

- In production, a model can achieve high accuracy but fail latency SLOs, violate conformal coverage, or degrade under INT8 quantization.
- TaxonVision enforces a strict **Four-Gate Promotion Verdict** in Stage 7 (`evaluate.py`) before any model receives the `@Production` tag.

The Four Automated Promotion Gates

Gate 1: Predictive Superiority

- Validation Macro PR-AUC $\geq$ active production baseline.

Gate 2: Quantization Fidelity

- INT8 accuracy drop $\leq$ 3.0% relative to FP32 baseline.

Gate 3: Empirical Conformal Coverage

- Holdout test coverage $\geq$ 90.0% (nominal 95%).

Gate 4: Latency Ceiling & Exchangeability

- CPU p95 latency $\leq$ 120 ms and live MMD $p \geq 0.01$.

Summary: The Three Pillars of TaxonVision MLOps

Robust Foundations, Epistemic Safety, and Closed-Loop Automation

1. Robust MLOps

- 7-Stage DVC Pipeline
- Dual-Tier S3 Remote
- MLflow 3.x Registry
- Zero-Copy NVMe Cache
- Pure Pixi Toolchain

2. Speed & Safety

- Sub-15ms INT8 ONNX
- 95% Coverage Guarantee
- Helmholtz Energy Filter
- Unified FastAPI/HTMX
- Forward Grad-CAM CAM

3. Closed Loop

- Postgres 384-D Ingestion
- Evidently 0.7 Drift
- Grafana Webhook Relay
- Label Studio Active Learn
- 4-Gate Auto Retraining

Key Takeaway: Production MLOps is not about training a model once in a notebook; it is about building a self-monitoring, self-healing, mathematically safe engineering loop.

Live Demonstration & Q&A

Real-Time Inference · Conformal Sets · Observability Cockpit

1. HTMX UI & CAM
2. Grafana Cockpit
3. MLflow Registry
4. Airflow Retraining

Thank you for your attention!

We now transition to the live system demonstration and jury questions.